

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Course: B. Tech. Final Year Branch : Electrical & Instrumentation Engineering

Semester :VII Subject Code & Name: BTEIEC703/High Voltage Engineering

Marks: 60

Date: 01/02/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Define electric field stress. Derive Poisson's equation.	Evaluation	6
B) Explain the estimation and control of electric stress.	Understand	6
C) What do you mean by surge voltages? How they are distributed and controlled?	Remember	6
Q.2 Solve Any Two of the following.		12
A) Explain Townsend's criterion for breakdown in gases.	Evaluation	6
B) State and explain Paschen's law.	Evaluation	6
C) Write a note on glow and arc discharge.	Remember	6
Q.3 Solve Any Two of the following.		12
A) Explain breakdown in commercial liquids.	Understand	6
B) What is 'Thermal breakdown' in solid dielectrics, and how is it practically more significant than other mechanisms?	Understand	6
C) Explain application of insulating materials in rotating machines.	Remember	6
Q.4 Solve Any Two of the following.		12
A) Explain propagation of lightning voltage and current waves on transmission lines.	Evaluation	6
B) Explain the concept of insulation coordination on high voltage power systems.	Understand	6
C) Explain with a neat sketch, surge diverter with its function.	Remember	6

Q.5	Solve Any Two of the following.		12
A)	Explain in detail Van de Graff generator with a neat sketch.	Remember	6
B)	Explain the measurement of high a.c. voltage using capacitive voltage transformer.	Understand	6
C)	Explain the testing of pin type insulators and bushings.	Remember	6

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